DATA SCIENCE 101

Introduction to Data Science

What data science is, why it matters, and how we start using data — from raw facts to machine learning.

SO Vengleab

TODAY'S AGENDA

What will we cover?

1

What

What **data**, **information**, and **data science** actually are — and how they differ.

2

Why

Why data science is **so popular** right now, and what data can **do for an organization**.

3

How

How we **start using data**: collect, prepare, explore, predict

Insight: By the end of this course, you'll have a better grasp of how data is being used around you — and how you can use it.

Question to Google

Data VS Information

What Is Data?

	lon	lat	alt	time
0	100.491491	13.725929	9.8	2018-07-24 07:32:31
1	100.491541	13.725939	10.5	2018-07-24 07:32:33
2	100.491575	13.725943	10.7	2018-07-24 07:32:35
3	100.491629	13.725970	10.8	2018-07-24 07:32:37
4	100.491667	13.725987	11.1	2018-07-24 07:32:38
5	100.491693	13.726010	11.0	2018-07-24 07:32:39
6	100.491723	13.726035	10.9	2018-07-24 07:32:40
7	100.491756	13.726069	10.8	2018-07-24 07:32:41
8	100.491801	13.726124	10.6	2018-07-24 07:32:42
9	100.491866	13.726193	10.6	2018-07-24 07:32:43

What is Information?

FOUNDATIONS

What Is Data?

Data = raw, unorganized facts.

- On its own, data is just **recorded values** — numbers, timestamps, coordinates.
- The table shows raw GPS readings: **lat**, **alt**, **time** — accurate, but meaningless until processed.
- Data needs **context and structure** before it can tell us anything.

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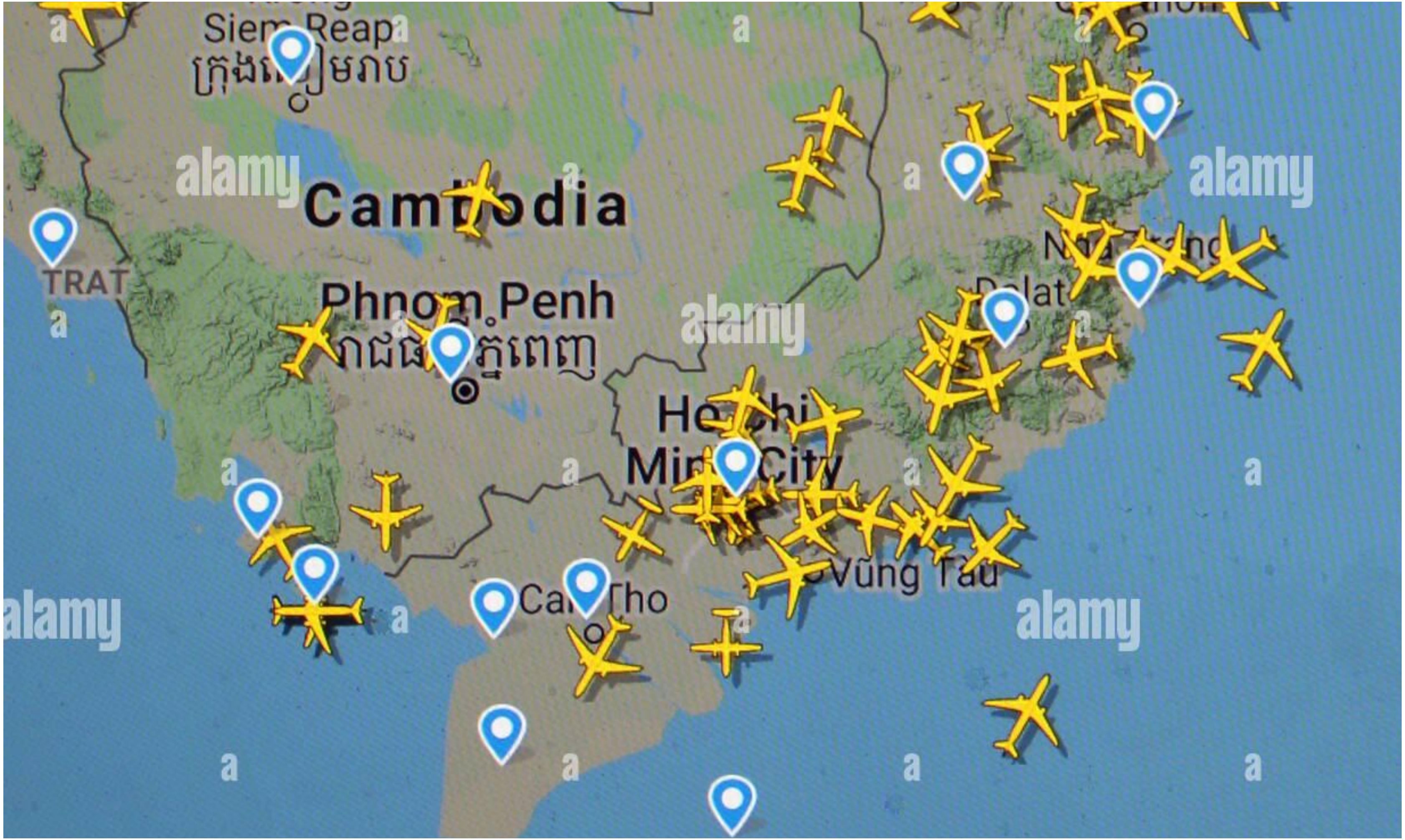
FOUNDATIONS

From Data to Information

Information = data with context.

- Plot those same GPS points on a map and suddenly they show **routes, places, and movement**.
- When data is **organized and presented in context**, it becomes information we can act on.

What is Data Science?


Siem Reap
Cambodia
TRAT
Phnom Penh
Ho Chi Minh City
Dalat
Nha Trang
Vũng Tàu
Can Tho

DEFINITION

What is Data Science?

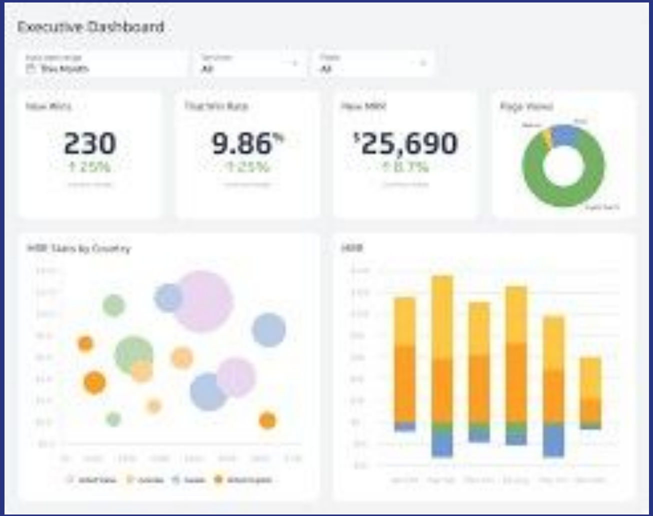
- **Data science** is a set of methodologies for taking the thousands of forms of data available to us today and using them to **draw meaningful conclusions**.
- We use data to better **describe the present** or **predict the future**.

Insight: If it helps you describe or predict, it's data science — regardless of the tool.

Describe the Current State

Describe a process as it is right now.

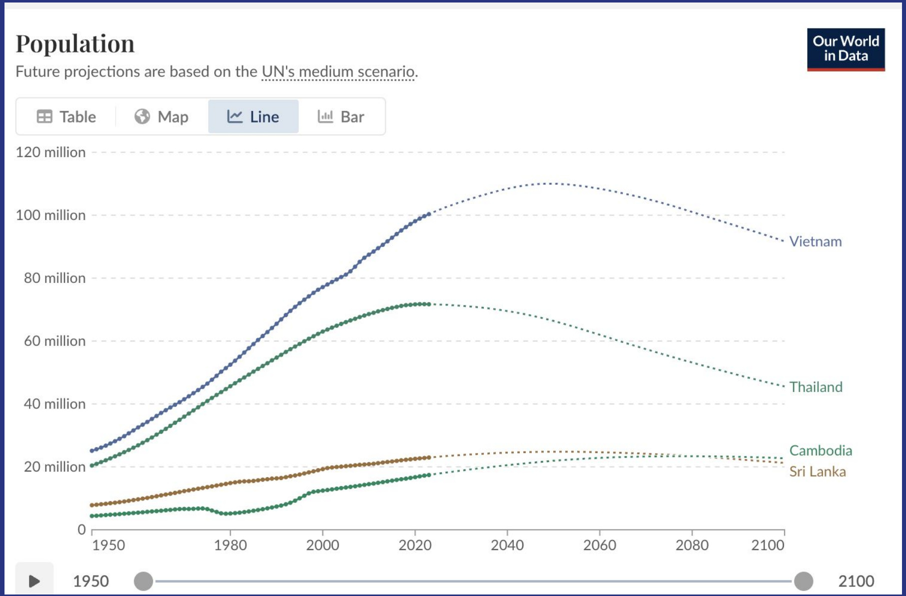
- Example: tracking **energy consumption** across buildings or teams.
- Dashboards and reports turn raw meter readings into a **clear picture of today**.



Predict Future Events

Forecast what happens next.

- Example: **forecasting population size** (see ourworldindata.org/population-growth).
- Models take **multiple causes into account**, predict potential outcomes, and let us evaluate the **probability of each prediction**.



Detect Anomalous Events

Spot abnormal events automatically.

- Example: flagging **fraudulent purchases** the moment they occur.
- Automatic anomaly detection **increases efficiency**: machines watch every event so people only review the abnormal ones.

Diagnose Causes

Explain WHY events and behaviors happen.

- For instance, the activity on **Spotify** or **Netflix**: what actually drives listening and watching?
- Diagnosis goes beyond **correlations between small numbers of events** to find the underlying causes.

WHY NOW

Why Is Data Science So Popular?

We're collecting more data than ever.

- **Data is everywhere**: every purchase, click, sensor, and message generates it.
- Storage is cheap, compute is powerful, and organizations that use their data **outperform those that don't**.

HOW WE USE DATA — STEPS 1–2

Collect, Store, and Prepare

Step 1 — Collection & Storage.

- Sources: **surveys**, **web traffic**, **geo-tagged social posts**, **financial transactions, POS system**

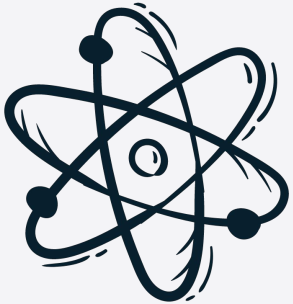
Step 2 — Data Preparation:

- cleaning, finding **missing or duplicate values**, converting data into an **organized format**.

Explore, Then Experiment

Step 3 — Explore & visualize (EDA).

- Build **dashboards** to track how data changes, or run **comparisons between sets of data**.
- Or **train & build ai and machine learning models** to predict events



Step 4 — Experimentation & Prediction:

- e.g., an **A/B test** to find which web page acquires more customers.

REVIEW LESSON

Check Your Understanding

1

Definitions

What is **data**? What is **information**? What is **data science**?

2

Differences

What is the **difference** between data, information, and data science?

3

Application

Why do we need data? **How** do we do data science?